Software Requirements Specification (SRS)

Project	IT Infrastructure & Asset Management System
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Revision History

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1.0	17-10-2026	SRI RAM V, Sharvesh, Yashwanth, Shreyan Ganesh	Initial SRS for IT Infrastructure & Asset Management System	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	To be assigned		
Project Guide	To be assigned		

Table of Contents

- 1. Introduction
- 2. Overall Description
- 3. External Interface Requirements
- 4. System Features and Functional Requirements
- 5. Non-functional Requirements
- 6. Security Objectives and Requirements
- 7. Quality Attributes and Acceptance Tests
- 8. UML Use-Case Model
- 9. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for an IT Infrastructure & Asset Management System (ITAMS). It establishes the expected behaviour, interfaces, constraints, security controls and acceptance criteria for tracking organizational hardware, software, licenses, warranties, maintenance and asset allocation.

1.2 Scope

The system provides a centralized inventory and lifecycle record for IT assets. Authorized users can register and update assets, allocate and return equipment, maintain software and license records, track warranty and maintenance dates, record service events, view compliance status, and receive alerts for expiry, maintenance and compliance issues. The initial

academic implementation assumes one organization and role-based access. Procurement, financial accounting, remote device management and automatic discovery from endpoint agents are outside the baseline scope unless added later.

1.3 Audience

Project team, faculty evaluator, system developers, testers, IT administrators, asset managers, department managers and compliance/audit personnel.

1.4 Definitions

Asset: an identifiable hardware or software item owned or managed by the organization. Asset tag: unique inventory identifier. CMDB: Configuration Management Database. RBAC: Role-Based Access Control. SLA: Service-Level Agreement. RTM: Requirements Traceability Matrix. MFA: Multi-Factor Authentication. PII: Personally Identifiable Information.

2. Overall Description

2.1 Product Perspective

ITAMS is a web-based application with a presentation layer, application/API layer and relational database. Users access it through a browser. The application stores asset master records, assignments, license entitlements, warranty dates, maintenance history, alerts and audit events. Email or in-app notifications may be used depending on deployment configuration.

2.2 Major Product Functions

- Register, edit, search, filter and retire hardware and software assets.
- Assign assets to employees, teams, departments or locations; record returns and transfers.
- Maintain serial number, asset tag, category, manufacturer, model, purchase date, status and location.
- Track software products, license keys or references, entitlement quantity, expiry and renewal status without exposing secrets in logs.
- Record warranty provider, start/end dates and support reference.
- Schedule preventive maintenance and record incidents, service actions, cost and next due date.
- Generate alerts for warranty/license expiry, overdue maintenance, missing fields and policy violations.
- Provide dashboards and reports for inventory, allocation, lifecycle status, upcoming expiry and compliance.
- Authenticate users and enforce role-based permissions.
- Maintain audit history for important changes and support CSV/PDF report export where enabled.

2.3 User Roles and Characteristics

Role	Responsibilities / Permissions
System Administrator	Manage users, roles, configuration, categories and all asset records.
Asset Manager / IT Admin	Create and update assets, allocate/return assets, manage licenses, warranties and maintenance.
Department Manager	View assets assigned to their department and approve allocation/transfer requests if enabled.
Employee / Asset Holder	View assigned assets and submit return, fault or maintenance requests.
Auditor / Read-only User	View reports, compliance status and audit history without modifying records.

2.4 Operating Environment

Modern desktop or mobile browser; application server running in an approved organizational or academic environment; relational database such as PostgreSQL or MySQL; network connectivity; optional SMTP or notification service. The implementation may be deployed locally for demonstration.

2.5 Constraints and Assumptions

- Each asset must have a unique system ID; asset tags and serial numbers should be unique when provided.
- Only authorized roles may change asset ownership, license details or compliance records.
- Date-based alerts depend on accurate date/time configuration and valid expiry/due dates.
- The organization supplies valid asset and employee data; imported data must be reviewed for duplicates and errors.
- Sensitive license credentials should be stored only if necessary and protected using appropriate secret-management controls; the baseline should prefer masked references.
- External email, HR, procurement and endpoint-discovery integrations are optional and may be represented by test stubs.

3. External Interface Requirements

3.1 User Interface

The interface shall provide a login page, dashboard, asset inventory list, asset details form, allocation/return screens, license and warranty views, maintenance calendar, alert centre, reports and administration screens. Forms shall show validation errors near the relevant fields. Search and filtering should support asset tag, name, category, status, department and assigned user.

3.2 Hardware Interfaces

No direct hardware interface is mandatory for the baseline web application. The system records details about organizational devices but does not control them. Barcode/QR scanning may be supported as keyboard input from a scanner if available.

3.3 Software Interfaces

- Relational database accessed through a parameterized data-access layer or ORM.
- Optional SMTP service for email notifications; failures must be logged without exposing sensitive values.
- Optional CSV import/export for migration and reporting.
- Optional identity provider/SSO integration; local authentication is acceptable for the academic prototype.

3.4 Communications

Production deployment should use HTTPS/TLS for browser-to-server traffic. API requests should use authenticated sessions or tokens. Database access must be restricted to the application service. Notification retries should be bounded and duplicate alerts avoided.

4. System Features and Functional Requirements

Requirement IDs use ITAM-F-###. Unless stated otherwise, requirements are high priority and are verified through the associated acceptance test.

Req ID	Requirement (shall)	Priority	Test Ref	Acceptance Criteria
ITAM-F-001	The system shall authenticate users before allowing access to protected functions.	High	TC-AUTH-01	Valid credentials open an authorized session; invalid credentials are rejected.
ITAM-F-002	The system shall enforce role-based permissions for administrators, asset managers, department managers, employees and auditors.	High	TC-SEC-01	Users cannot perform actions outside their assigned role.
ITAM-F-003	The system shall create a unique asset record with a system-generated asset ID.	High	TC-ASSET-01	A saved asset receives a unique ID and is retrievable.
ITAM-F-004	The system shall store core hardware/software asset fields including name, category, tag, serial number where applicable, status and	High	TC-ASSET-02	Valid fields persist and appear on the asset details page.

	location.			
ITAM-F-005	The system shall validate required fields and warn about duplicate asset tags or serial numbers where supplied.	High	TC-ASSET-03	Missing required fields and duplicate identifiers are rejected or flagged.
ITAM-F-006	The system shall allow authorized users to search, filter and sort asset records.	High	TC-SEARCH-01	Search results match the selected filters.
ITAM-F-007	The system shall allocate an available asset to a user, team, department or location and record allocation date and responsible user.	High	TC-ALLOC-01	Allocation history and current holder are updated.
ITAM-F-008	The system shall record asset return, reassignment and transfer history.	High	TC-ALLOC-02	Return/transfer updates status and preserves prior history.
ITAM-F-009	The system shall maintain software product and license records including entitlement quantity, expiry date and renewal status.	High	TC-LIC-01	License data is saved and visible to authorized users.
ITAM-F-010	The system shall flag license usage that exceeds recorded entitlement quantity when usage data is available.	High	TC-COMP-01	An over-allocation is flagged in the compliance view.
ITAM-F-011	The system shall record warranty provider/reference, start date, end date and support contact where available.	High	TC-WAR-01	Warranty details are stored and displayed correctly.
ITAM-F-012	The system shall create maintenance schedules with due date, priority, status and responsible technician/team.	High	TC-MAINT-01	A scheduled maintenance item appears in the maintenance list/calendar.
ITAM-F-013	The system shall record maintenance and repair history against the related asset.	High	TC-MAINT-02	A completed service event is linked to the asset with date and notes.
ITAM-F-014	The system shall generate in-app alerts for approaching or passed license and warranty expiry dates.	High	TC-ALERT-01	Configured date thresholds create the correct alert.
ITAM-F-015	The system shall generate alerts for overdue maintenance and missing mandatory asset/compliance information.	High	TC-ALERT-02	Overdue and incomplete records appear in the alert centre.
ITAM-F-016	The system shall allow authorized users to acknowledge, assign and resolve alerts while retaining status history.	Medium	TC-ALERT-03	Alert state changes are timestamped and auditable.
ITAM-F-017	The system shall provide dashboard summaries for asset counts by category, status, department and lifecycle state.	Medium	TC-DASH-01	Dashboard totals match the underlying test dataset.
ITAM-F-018	The system shall generate inventory, allocation, maintenance, expiry and compliance reports.	High	TC-REPORT-01	Report content matches filters and source records.
ITAM-F-019	The system shall export authorized report results to CSV or PDF where the selected export format is enabled.	Medium	TC-REPORT-02	Export opens successfully and contains the selected data.
ITAM-F-020	The system shall maintain timestamped audit events for asset creation, modification, allocation, return, license changes and administrative	High	TC-AUDIT-01	Audit records identify action, actor, time and affected record.

	actions.			
ITAM-F-021	The system shall support deactivation/retirement of assets without deleting historical allocation and maintenance records.	High	TC-LIFE-01	Retired assets are excluded from available stock but remain in reports/history.
ITAM-F-022	The system shall allow authorized users to import asset data from a validated CSV template and report rejected rows.	Medium	TC-IMPORT-01	Valid rows import; invalid rows return row-level errors without silent data loss.

5. Non-functional Requirements

Req ID	Requirement	Category	Priority	Acceptance / Measurement
ITAM-NF-001	Under normal test load, 95% of common list/search requests shall complete within 3 seconds for a dataset of 10,000 assets.	Performance	High	Load test: p95 ≤ 3 s.
ITAM-NF-002	The application shall target 99.5% monthly availability excluding planned maintenance for a production deployment.	Reliability	Medium	Availability report meets target.
ITAM-NF-003	The application shall validate all user input server-side and use parameterized database operations.	Security	High	Input/security tests show no injection vulnerability.
ITAM-NF-004	The system shall maintain a complete audit trail for designated business-critical changes and retain it according to organizational policy.	Auditability	High	Audit test verifies actor, timestamp, action and record.
ITAM-NF-005	The UI shall support current versions of major browsers and provide clear labels, keyboard navigation and readable validation messages.	Usability/Accessibility	Medium	Manual browser and accessibility checklist passes.
ITAM-NF-006	The system shall protect stored and transmitted sensitive data using appropriate encryption and access controls.	Confidentiality	High	Configuration review and security tests pass.
ITAM-NF-007	The application shall provide database backup and documented restoration procedures for the chosen deployment.	Recoverability	Medium	Restore test recovers sample records within the agreed recovery objective.
ITAM-NF-008	The application shall use modular components and documented APIs so inventory, allocation, alerting and reporting can be maintained independently.	Maintainability	Medium	Code review confirms module boundaries and documentation.

6. Security Objectives and Requirements

6.1 Security Objectives

- Confidentiality: prevent unauthorized access to employee assignments, license information and administrative records.
- Integrity and accountability: ensure asset, allocation and compliance records are changed only by authorized users and important changes can be audited.
- Availability: keep inventory and maintenance information accessible to authorized staff and recover it after a service or data failure.

6.2 Security Requirements

Req ID	Security Requirement	Priority	Acceptance Test
ITAM-SR-001	All browser-to-server traffic in deployed environments shall use HTTPS/TLS.	High	TLS configuration scan passes.
ITAM-SR-002	The system shall enforce server-side RBAC checks for every protected operation.	High	Negative authorization tests are denied.
ITAM-SR-003	Passwords shall be stored using a suitable salted password-hashing algorithm and shall never be stored in plaintext.	High	Database inspection confirms hashes only.
ITAM-SR-004	The system shall not write passwords, session tokens or full license secrets to application logs.	High	Log review finds no secrets.
ITAM-SR-005	The system shall record audit events for privileged and lifecycle-changing operations and restrict audit-log modification to authorized administrators.	High	Audit integrity/permission tests pass.
ITAM-SR-006	The system shall validate and normalize input and use parameterized queries to reduce injection and malformed-input risks.	High	Security test payloads are safely rejected/handled.

7. Quality Attributes and Acceptance Tests

- Functional completeness: all high-priority functional requirements are implemented and pass their mapped tests.
- Security: no open critical/high security defects at acceptance; authorization, input validation and secret-handling tests pass.
- Performance: ITAM-NF-001 is met in the defined test environment and dataset.
- Data integrity: asset IDs are unique and allocation, maintenance and audit history remain consistent after updates.
- Usability: representative users can find an asset, allocate it, review an expiry alert and generate a report using the documented UI.
- Acceptance exit: planned tests executed, critical defects closed, high-priority requirements passed or formally waived, and RTM status updated.

8. UML Use-Case Model

Primary actors: System Administrator, Asset Manager/IT Admin, Department Manager, Employee/Asset Holder and Auditor. Main use cases: authenticate, manage users/roles, register/update/retire asset, search inventory, allocate/return/transfer asset, manage software license, manage warranty, schedule/record maintenance, review/resolve alerts, view dashboard, generate/export reports and inspect audit history.

Use-case diagram specification (renderable in PlantUML):

```
@startuml
left to right direction
actor "System Administrator" as Admin
actor "Asset Manager / IT Admin" as Manager
actor "Department Manager" as Dept
actor "Employee / Asset Holder" as Employee
actor Auditor
rectangle ITAMS {
  usecase "Authenticate" as U1
  usecase "Manage Users and Roles" as U2
  usecase "Manage Asset Inventory" as U3
  usecase "Allocate / Return Asset" as U4
  usecase "Manage Licenses and Warranties" as U5
  usecase "Schedule / Record Maintenance" as U6
  usecase "Review and Resolve Alerts" as U7
  usecase "View Dashboard and Reports" as U8
  usecase "View Audit History" as U9
}
Admin --> U1
Admin --> U2
Admin --> U3
Admin --> U5
Admin --> U6
Admin --> U7
Admin --> U8
Admin --> U9
Manager --> U1
Manager --> U3
Manager --> U4
Manager --> U5
Manager --> U6
Manager --> U7
Manager --> U8
Dept --> U1
Dept --> U4
Dept --> U8
Employee --> U1
Employee --> U4
Auditor --> U1
Auditor --> U8
Auditor --> U9
@enduml
```

9. Requirements Traceability Matrix (RTM)

Req ID	Requirement Summary	Module	Verification	Status
ITAM-F-001	The system shall authenticate users before allowing acc...	Authentication	TC-AUTH-01	Not run
ITAM-F-002	The system shall enforce role-based permissions for adm...	Authorization	TC-SEC-01	Not run
ITAM-F-003	The system shall create a unique asset record with a sy...	Asset Inventory	TC-ASSET-01	Not run
ITAM-F-004	The system shall store core hardware/software asset fie...	Asset Inventory	TC-ASSET-02	Not run
ITAM-F-005	The system shall validate required fields and warn abou...	Validation	TC-ASSET-03	Not run
ITAM-F-006	The system shall allow authorized users to search, filt...	Search	TC-SEARCH-01	Not run
ITAM-F-007	The system shall allocate an	Allocation	TC-ALLOC-01	Not run

	available asset to a user,...			
ITAM-F-008	The system shall record asset return, reassignment and ...	Allocation	TC-ALLOC-02	Not run
ITAM-F-009	The system shall maintain software product and license ...	Licenses	TC-LIC-01	Not run
ITAM-F-010	The system shall flag license usage that exceeds record...	Compliance	TC-COMP-01	Not run
ITAM-F-011	The system shall record warranty provider/reference, st...	Warranty	TC-WAR-01	Not run
ITAM-F-012	The system shall create maintenance schedules with due ...	Maintenance	TC-MAINT-01	Not run
ITAM-F-013	The system shall record maintenance and repair history ...	Maintenance	TC-MAINT-02	Not run
ITAM-F-014	The system shall generate in-app alerts for approaching...	Alerts	TC-ALERT-01	Not run
ITAM-F-015	The system shall generate alerts for overdue maintenanc...	Alerts	TC-ALERT-02	Not run
ITAM-F-016	The system shall allow authorized users to acknowledge,...	Alerts	TC-ALERT-03	Not run
ITAM-F-017	The system shall provide dashboard summaries for asset ...	Dashboard	TC-DASH-01	Not run
ITAM-F-018	The system shall generate inventory, allocation, mainte...	Reporting	TC-REPORT-01	Not run
ITAM-F-019	The system shall export authorized report results to CS...	Export	TC-REPORT-02	Not run
ITAM-F-020	The system shall maintain timestamped audit events for ...	Audit	TC-AUDIT-01	Not run
ITAM-F-021	The system shall support deactivation/retirement of ass...	Lifecycle	TC-LIFE-01	Not run
ITAM-F-022	The system shall allow authorized users to import asset...	Import	TC-IMPORT-01	Not run
ITAM-NF-001	Under normal test load, 95% of common list/search reque...	Performance	Load test: p95 ≤ 3 s	Not run
ITAM-NF-002	The application shall target 99.5% monthly availability...	Reliability	Availability report meets target	Not run
ITAM-NF-003	The application shall validate all user input server-si...	Security	Input/security tests show no injection vulnerability	Not run
ITAM-NF-004	The system shall maintain a complete audit trail for de...	Auditability	Audit test verifies actor, timestamp, action and record	Not run
ITAM-NF-005	The UI shall support current versions of major browsers...	Usability/Accessibility	Manual browser and accessibility checklist passes	Not run
ITAM-NF-006	The system shall protect stored and transmitted sensiti...	Confidentiality	Configuration review and security tests pass	Not run
ITAM-NF-007	The application shall provide database backup and docum...	Recoverability	Restore test recovers sample records within the agreed recovery objective	Not run
ITAM-NF-008	The application shall use modular components and docume...	Maintainability	Code review confirms module boundaries and documentation	Not run
ITAM-SR-001	All browser-to-server traffic in deployed environments ...	Security	TLS configuration scan passes.	Not run
ITAM-SR-002	The system shall enforce server-side RBAC checks for	Security	Negative authorization tests are denied.	Not run

	ev...			
ITAM-SR-003	Passwords shall be stored using a suitable salted passw...	Security	Database inspection confirms hashes only.	Not run
ITAM-SR-004	The system shall not write passwords, session tokens or...	Security	Log review finds no secrets.	Not run
ITAM-SR-005	The system shall record audit events for privileged and...	Security	Audit integrity/permission tests pass.	Not run
ITAM-SR-006	The system shall validate and normalize input and use p...	Security	Security test payloads are safely rejected/handled.	Not run